NLP 201: Large Language Models

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Slides adapted from Mohit Iyyer

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Plan

- Prompting and In-Context Learning
- Retrieval-Augmented Methods
- Instruction Tuning

The language model “scaling wars”!

ELMo: 93M params, 2-layer biLSTM

BERT-base: 110M params, 12-layer Transformer

BERT-large: 340M params, 24-layer Transformer

Model Name	n_{params}	n_{layers}	d_{model}	n_{heads}	d_{head}	Batch Size	Learning Rate
GPT-3 Small	125M	12	768	12	64	0.5M	6.0×10^{-4}
GPT-3 Medium	350M	24	1024	16	64	0.5M	3.0×10^{-4}
GPT-3 Large	760M	24	1536	16	96	0.5M	2.5×10^{-4}
GPT-3 XL	1.3B	24	2048	24	128	1M	2.0×10^{-4}
GPT-3 2.7B	2.7B	32	2560	32	80	1M	1.6×10^{-4}
GPT-3 6.7B	6.7B	32	4096	32	128	2M	1.2×10^{-4}
GPT-3 13B	13.0B	40	5140	40	128	2M	1.0×10^{-4}
GPT-3 175B or “GPT-3”	175.0B	96	12288	96	128	3.2M	0.6×10^{-4}

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The language model “scaling wars”!

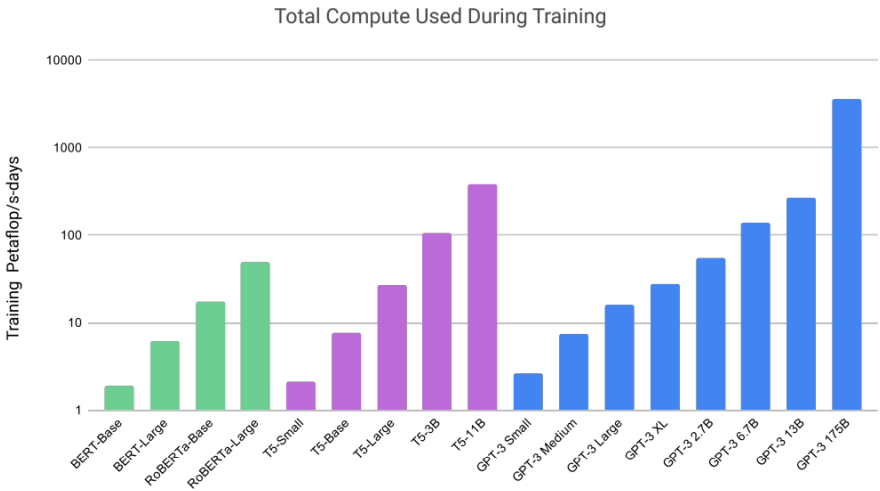
ELMo: 1B training tokens

BERT: 3.3B training tokens

RoBERTa: ~30B training tokens

Dataset	Quantity (tokens)	Weight in training mix	Epochs elapsed when training for 300B tokens
Common Crawl (filtered)	410 billion	60%	0.44
WebText2	19 billion	22%	2.9
Books1	12 billion	8%	1.9
Books2	55 billion	8%	0.43
Wikipedia	3 billion	3%	3.4

The language model “scaling wars”!



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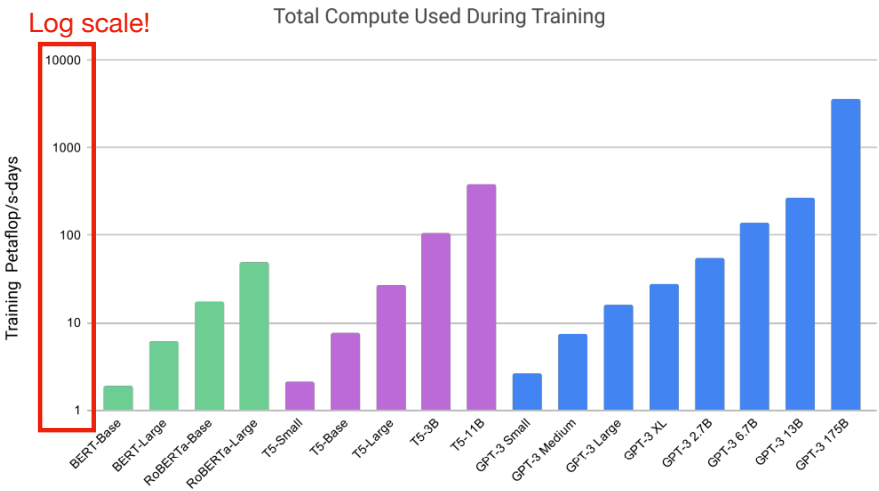


2022-2023:

PaLM (Google): 540B params, 118 layers, 18432 d_model, 780 billion training tokens Model not available

ChatGPT (OpenAI): Params, layers, dimensionality, training data size unknown Model available only through blackbox API

LLaMa (Meta): 65B params, 80 layers, 8192 d_model, 1.4 trillion tokens of training data Model parameters publicly available!

GPT4 (OpenAI): Params, layers, dimensionality, training data size unknown Model available only through blackbox API

Bard (Google): Params, layers, dimensionality, training data size unknown Model available only through blackbox API

so… what does all of this scaling buy us?

Traditional fine-tuning (not used for GPT-3)

Fine-tuning

The model is trained via repeated gradient updates using a large corpus of example tasks.

Downstream training data

```
sea otter => loutre de mer
```

example #1

gradient update

```
peppermint => menthe poivrée
```

example #2

gradient update

...

```
plush giraffe => girafe peluche
```

example #N

gradient update

```
cheese =>
```

prompt

Downstream test data

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
Translate English to French:
cheese =>
```

task description

prompt

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
Translate English to French:
cheese =>
```

task description

prompt

No fine-tuning!!! Literally just take a pretrained LM and give it the following prefix:

“Translate English to French: cheese =>”

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
Translate English to French:        ← task description
sea otter => loutre de mer          ← example
cheese =>                           ← prompt
```

No fine-tuning!!! Literally just take a pretrained LM and give it the following prefix:

“Translate English to French: sea otter => loutre de mer, cheese =>”

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1  Translate English to French:        ← task description
2  sea otter => loutre de mer          ← examples
3  peppermint => menthe poivrée        ←
4  plush girafe => girafe peluche      ←
5  cheese =>                           ← prompt
```

No fine-tuning!!! Literally just take a pretrained LM and give it the following prefix:

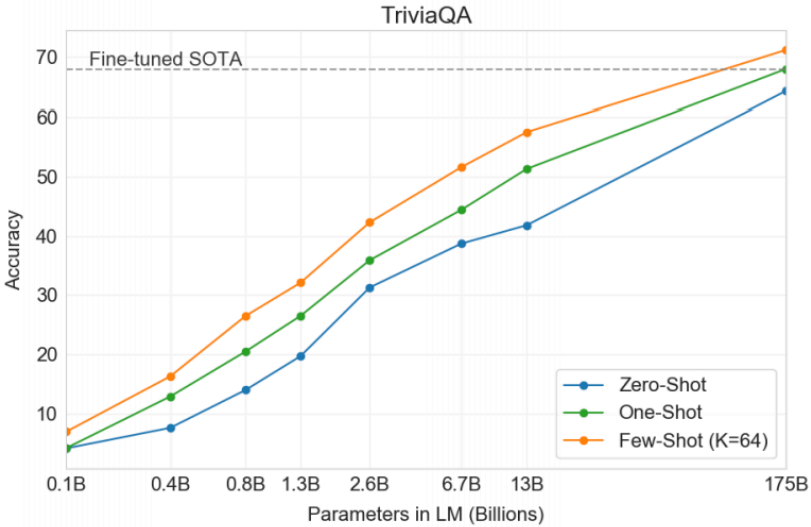
“Translate English to French: sea otter => loutre de mer, peppermint => … (few more examples), cheese =>”

Max of 100 examples fed into the prefix in this way

How does this new paradigm compare to “pretrain + finetune”?

TriviaQA

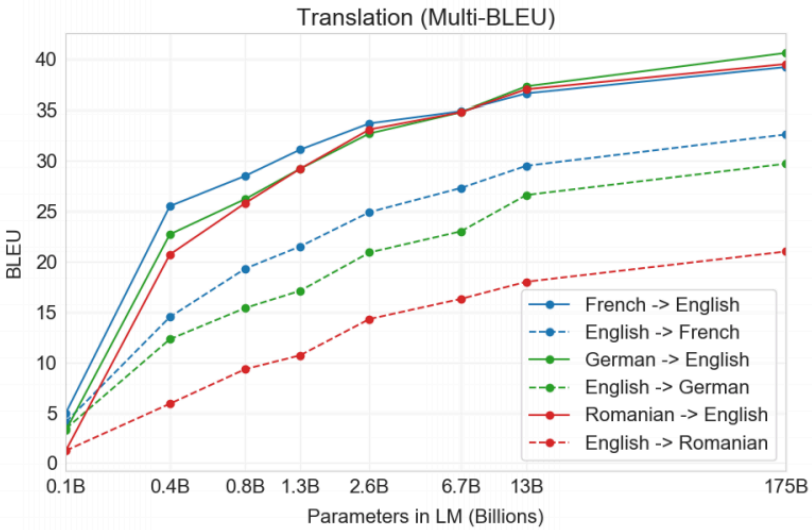
Question
Miami Beach in Florida borders which ocean?
What was the occupation of Lovely Rita according to the song by the Beatles
Who was Poopdeck Pappys most famous son?
The Nazi regime was Germany's Third Reich; which was the first Reich?
At which English racecourse did two horses collapse and die in the parade ring due to electrocution, in February 2011?
Which type of hat takes its name from an 1894 novel by George Du Maurier where the title character has the surname O'Ferrall ?
What was the Elephant Man's real name?


TriviaQA
Fine-tuned SOTA
Accuracy
70
60
50
40
30
20
10
0.1B
0.4B
0.8B
1.3B
2.6B
6.7B
13B
175B
Parameters in LM (Billions)
Zero-Shot
One-Shot
Few-Shot (K=64)

What about translation? (7% of GPT3's training data is in languages other than English)

Setting	En→Fr	Fr→En	En→De	De→En	En→Ro	Ro→En
SOTA (Supervised)	**45.6**[a]	35.0 [b]	**41.2**[c]	40.2[d]	**38.5**[e]	**39.9**[e]
XLM [LC19]	33.4	33.3	26.4	34.3	33.3	31.8
MASS [STQ+19]	37.5	34.9	28.3	35.2	35.2	33.1
mBART [LGG+20]	-	-	29.8	34.0	35.0	30.5
GPT-3 Zero-Shot	25.2	21.2	24.6	27.2	14.1	19.9
GPT-3 One-Shot	28.3	33.7	26.2	30.4	20.6	38.6
GPT-3 Few-Shot	32.6	39.2	29.7	40.6	21.0	39.5

Improvements haven't plateaued!

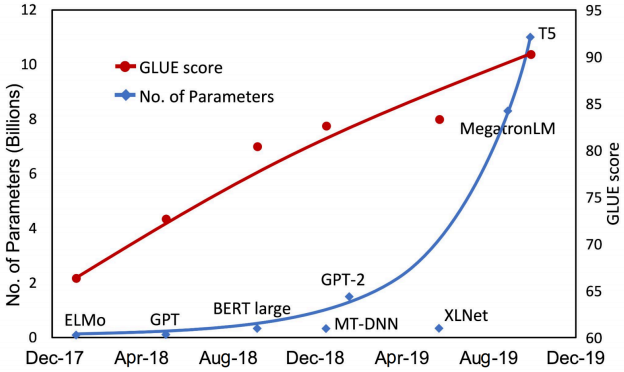


What about reading comprehension QA?

Setting	CoQA	DROP	QuAC	SQuADv2	RACE-h	RACE-m
Fine-tuned SOTA	**90.7**[a]	**89.1**[b]	**74.4**[c]	**93.0**[d]	**90.0**[e]	**93.1**[e]
GPT-3 Zero-Shot	81.5	23.6	41.5	59.5	45.5	58.4
GPT-3 One-Shot	84.0	34.3	43.3	65.4	45.9	57.4
GPT-3 Few-Shot	85.0	36.5	44.3	69.8	46.8	58.1

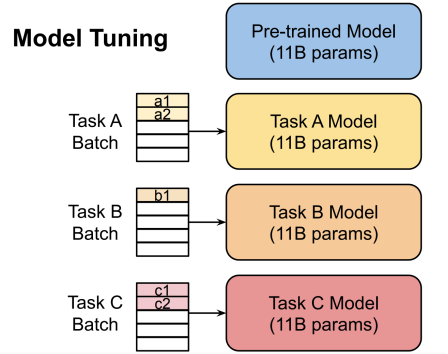
Struggles on “harder” datasets

Scaling up the model size is one of the most important ingredients for achieving the best performance



Ahmet and Abdullah., 2021

Practical challenges: large-scale models are costly to share and serve



Lester et al., 2021

Language model prompting to the rescue!

GPT-3 (Brown et al., 2020): In-context learning

- **natural language instruction** and/or **a few task demonstrations** ⟶ **output**

  "Translate English to German:" That is good ⟶ Das is gut

- *no* gradient updates or fine-tuning

Sub-optimal and sensitive discrete/hard prompts

Discrete/hard prompts

- natural language instructions/task descriptions

Problems

- requiring domain expertise/understanding of the model's inner workings
- performance still lags far behind SotA model tuning results
- sub-optimal and sensitive
  - prompts that humans consider reasonable is not necessarily effective for language models (Liu et al., 2021)
  - pre-trained language models are sensitive to the choice of prompts (Zhao et al., 2021)

Sub-optimal and sensitive discrete/hard prompts (cont.)

Prompt	P@1
[X] is located in [Y]. *(original)*	31.29
[X] is located in which country or state? [Y].	19.78
[X] is located in which country? [Y].	31.40
[X] is located in which country? In [Y].	51.08

Table 1. Case study on LAMA-TREx P17 with bert-base-cased. A single-word change in prompts could yield a drastic difference.

Liu et al., 2021

Shifting from discrete/hard to continuous/soft prompts

Progress in prompt-based learning

- manual prompt design (Brown et al., 2020; Schick and Schutze, 2021a,b)
- mining and paraphrasing based methods to automatically augment the prompt sets (Jiang et al., 2020)
- gradient-based search for improved discrete/hard prompts (Shin et al., 2020)
- automatic prompt generation using a separate generative language model (i.e., T5) (Gao et al., 2020)
- learning continuous/soft prompts (Liu et al., 2021; Li and Liang., 2021; Qin and Eisner., 2021; Lester et al., 2021)

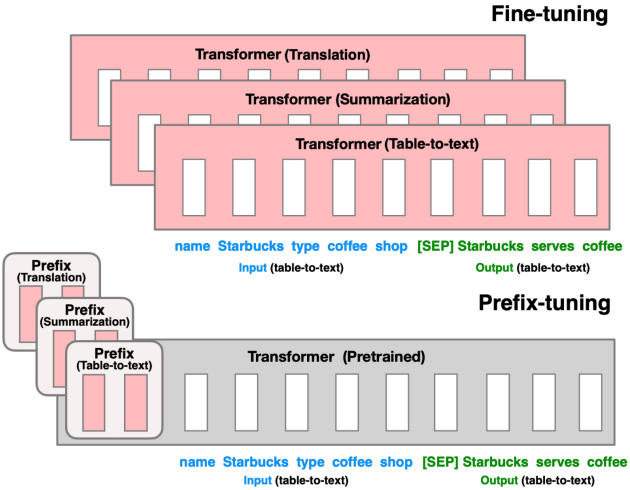
Continuous/soft prompts

- additional learnable parameters injected into the model

It remains unclear how to learn continuous/ soft prompts effectively?

- **P-tuning** (Liu et al., 2021): encode dependencies between prompt tokens using a BiLSTM network
- **P-tuning** (Liu et al., 2021), **Prefix Tuning** (Li and Liang., 2021): inject prompts at different positions of the input / model
- **P-tuning** (Liu et al., 2021): use mixed prompt initialization strategies
- **Soft Prompts** (Qin and Eisner., 2021): use ensemble methods, e.g., mixture-of-experts

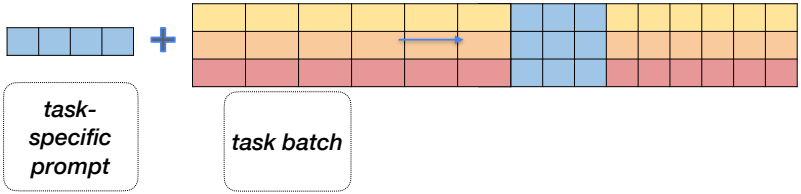
Prefix tuning (Li & Liang, ACL 2021)



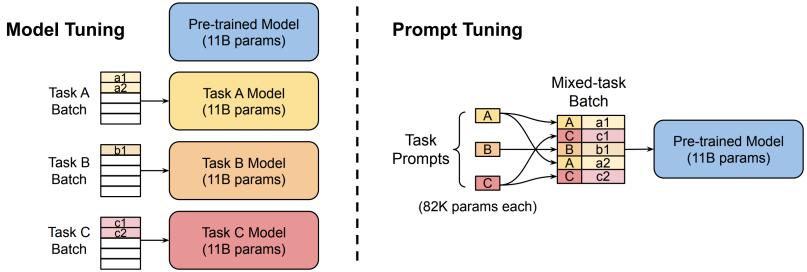
Prompt Tuning idea (Lester et al., 2021)

What is a prompt in Prompt Tuning?

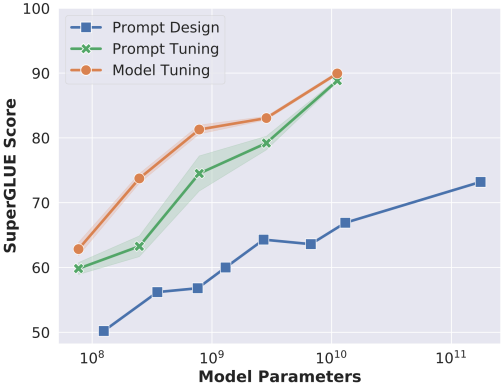
- a sequence of additional task-specific tunable tokens prepended to the input text

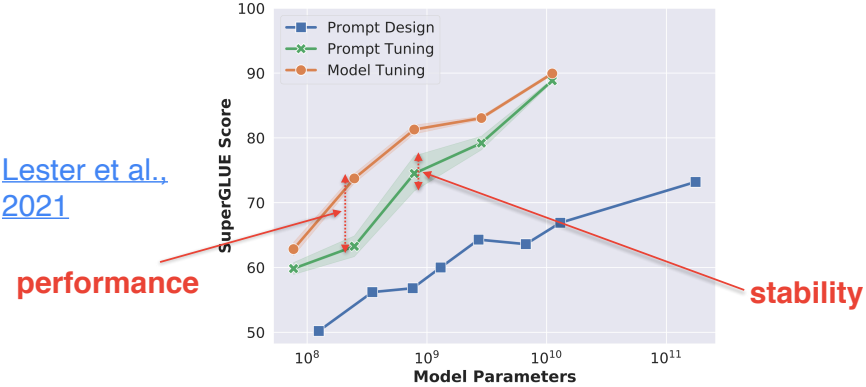


Parameter-efficient Prompt Tuning



Prompt Tuning becomes more competitive with scale



Room for improving Prompt Tuning

Lester et al., 2021